• DBMS (ENCS204) •

[ASSIGNMENT - 3]

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[B.tech CSE (core), Section- A]

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[UNIT - 3]

[Section- A]

Q1) what is SQL? why is it important in DBMS?

Ans SQL (structured query language) is the standard language used to communicate with relational Database Management Systems (RDBMS)

- It allows users to perform tasks such as creating tables, updating data, retrieving specific information with complex filters
- Managing databases - security / permissions.

Q2) what is the difference between DDL and DML?

Ans. • DDL (Data Definition Language) :- used to define or change the database structure (schema). Ex- create, ALTER, DROP.

- DML (Data Manipulation Language) :- used to manage and manipulate data within those structures. Ex:- INSERT, UPDATE, DELETE.

HAVING

Q3) what is a JOIN? Explain its types.

Ans A JOIN is used to combine rows from two or more tables based on a related coloumn between them.

- INNER JOIN :- Returns records that have matching values in both sides.

- LEFT (OUTER) JOIN :- Returns all records from the left table and matched records from the right.

- RIGHT (OUTER) JOIN :- Returns all records from the right table and matched records from the left.

- FULL (OUTER) JOIN :- Returns all records from when there is a match in either the left or right table.

Q4) what is a GROUP BY and HAVING clause?

- GROUP BY :- collapses multiple rows into summary rows. eg:- "find the number of students per city").
  - It is used with aggregate functions like COUNT, MAX, SUM.

- **HAVING**:- Acts a a filter for the groups created by GROUP BY. You cannot use WHERE with aggregate functions, so HAVING is used instead.

Q5) what is a Subquery?

- A Susquery is a query nested inside another SQL statement (the outer query) It is usually used in the WHERE or FROM clause to provide a data set that the outer query uses for further filtering or calculation.

Q6) what is relational algebra?

Ans It is a procedural query language that provides a theoritical foundation for relational databases. It uses operators like σ for select, π for project, and $\bowtie$ Join for Join) to perform operations on relations (tables) or produce new relations.

Q7) what is the difference between WHERE and HAVING?

Ans ~~filters~~ WHERE :- filters rows before any grouping is performed.

- HAVING :- filters groups after the GROUP BY clause has been applied.

[SECTION - B]

Q1) write an SQL query to retrieve students with marks >75.

Ans

```
SELECT * FROM Students WHERE
marks > 75;
```

Q2) W.A.Q using JOIN to combine Student and Course tables.

Ans

```
SELECT Student. Name,
Course. CourseName
FROM Student
INNER JOIN Course ON
Student. CourseID =
Course. CourseID;
```

Q3) W.A.Q using GROUP BY to count students per departure.

Ans

```
SELECT Department, COUNT(*) AS
Total Students
FROM Students
GROUP BY Department;
```

[SECTION- C]

Q1) A company database stores employee and department data. W.A.Q to find employees in a specific department.

Ans It requires a JOIN to connect the Employees table with the Departments table, followed by a WHERE clause to filter for the department name

Query:-

```
SELECT E. EmpName, D. DeptName
FROM Employees E
JOIN Departments D ON E. Dept ID
= D. Dept ID
WHERE D. DeptName =
'Engineering';
```

Q2) E - commerce System : Retrieve total sales per customer.

Ans . to use aggregate function (SUM) combined with GROUP BY clause to consolidate all individual orders belonging to a single customer ID.

Query.

```
SELECT Customer ID,
SUM (Total Amount) AS
Total Sales Value
FROM orders
GROUP BY Customer ID;
```

Q3) A university needs top 3 students. w.A.Q.

Ans To find "top" entries, you must sort the data in descending order (DESC) based on marks / GPA and then restrict the output to the first three rows.

Query :-

```
SELECT Student Name, GPA
FROM Students
ORDER BY GPA DESC
LIMIT 3;
```

[SECTION-D]

Q1) Write an SQL query for any real life system and explain Logic.

Ans System:- Hospital Management

Query:-
```
SELECT * FROM
Appointments WHERE
AppointmentDate = '2026-04-12';
```

Logic :- This allows the front desk to manage the daily schedule and resource allocation (doctors/rooms) for the upcoming day, ensuring patient flow is optimized.

Q2) Write a JOIN Query and mentione one use case.

Ans Query:-
```
SELECT Products. ProductName,
Categories. CategoryName,
FROM Products
INNER JOIN categories ON
Products. CategoryID =
Categories. CategoryID;
```

• Use Case :- This is essential for retail inventory system. By linking these tables, a manger can see exactly which products belong to which category.

Q31) ~~what~~ write a GROUP BY Query and explain its output.

Ans Query :-

```
SELECT City, COUNT (Store ID)
FROM Retail Stores,
GROUP BY City;
```

Explanation :- The output will provide a list of cities and the count of stores in each.